

Code Review Report

Plugin: qtype_buchungssatz

Plugin version: 2024010130

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Scope: Readiness for publication on moodle.org/plugins

Overall Verdict: NOT READY FOR PUBLICATION

The plugin contains 4 critical issues that must be resolved before submission to moodle.org/plugins. Additionally, 4 major issues and 5 minor issues require attention.

1. Issue Summary

CRITICAL 4 issues Block publication	MAJOR 4 issues Must be fixed	MINOR 5 issues Recommended
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2. Critical Issues

The following issues must be resolved before the plugin can be submitted to moodle.org/plugins.

2.1 Direct access to \$_FILES superglobal

File: edit_chart.php, line 145

The plugin reads the uploaded CSV file by accessing the PHP superglobal \$_FILES directly. This violates Moodle Security Guidelines, which prohibit direct use of superglobals in plugin code.

Required action: Replace with Moodle File API (get_file_storage()) or retrieve file content through \$mform->get_file_content().

2.2 Plugin maturity set to MATURITY_ALPHA

File: version.php

The `$plugin->maturity` value is currently set to `MATURITY_ALPHA`. While moodle.org/plugins technically accepts plugins with this maturity level, it explicitly signals that the plugin is not intended for production use. This is incompatible with a public release.

Required action: Set to MATURITY_BETA or MATURITY_STABLE before submission.

2.3 Incorrect Moodle version requirement

File: version.php

The `$plugin->requires` value is set to 2020110900, which corresponds to Moodle 3.10. The plugin targets Moodle 4.1 and above, as stated in the project documentation. This mismatch may cause the plugin to appear on incompatible installations.

Required action: Update to 2022112800 (Moodle 4.1).

2.4 AMD build directory missing

Directory: amd/build/

The `amd/build/` directory — which contains the compiled and minified JavaScript modules — is absent from the repository (gitignored). Moodle loads the minified files from this directory on production sites. Without them, all AMD-based interactivity will fail unless the site administrator has explicitly disabled JavaScript caching (`$CFG->cachejs = false`), which is not a standard production configuration.

Required action: Run grunt amd to compile the AMD modules and commit the resulting amd/build/ directory to the repository.

3. Major Issues

The following issues do not technically block submission but represent violations of Moodle Coding Standards or security guidelines that reviewers will flag.

3.1 Insecure license header URLs (20+ files)

All PHP and JavaScript files

All source files contain license headers with `http://` URLs pointing to the GNU license and Moodle documentation. Moodle Coding Style requires `https://` URLs. This affects more than 20 files.

Required action: Replace all http:// occurrences in license headers with https://.

3.2 Non-standard license header

All PHP and JavaScript files

The standard opening line for Moodle plugin files is:

```
// This file is part of Moodle - https://moodle.org/
```

The current files use a custom line referencing the plugin name instead. This will be flagged during the moodle.org review process.

Required action: Update the first line of the license header in all files to the standard Moodle format.

3.3 CSRF vulnerability: export action without sesskey check

File: edit_chart.php, line 170

The export action in edit_chart.php does not call confirm_sesskey() or require_sesskey(). All state-changing or data-exposing actions must verify the session key to prevent cross-site request forgery.

Required action: Add confirm_sesskey() at the beginning of the export action handler.

3.4 lib.php is empty

File: lib.php

lib.php contains only the file header. Moodle loads this file automatically for all question type plugins. An empty lib.php should either be removed or — if file serving is needed — implement the qtype_buchungssatz_pluginfile() callback.

Required action: Remove lib.php or implement the required callback function.

4. Minor Issues

4.1 Use of die; instead of exit

Files: manage_charts.php, edit_chart.php

Moodle Coding Style recommends using `exit` over `die`. While functionally equivalent, consistent use of `exit` is expected in code submitted to moodle.org.

4.2 Empty settings.php

File: settings.php

settings.php contains only the file header. Moodle automatically includes this file and will create an empty settings page for the plugin in Site Administration. If the plugin has no configurable settings, this file should be removed.

4.3 Legacy AMD module pattern

Files: amd/src/question.js, amd/src/editform.js

Both JavaScript files use the legacy AMD `define()` pattern with a jQuery dependency. Moodle 4.x is progressively moving toward ES modules (`import/export`). The legacy pattern still functions in Moodle 4.1, but new plugin submissions are encouraged to use native ES modules.

4.4 Missing README

A README file is required for moodle.org plugin submissions. It should include a plugin description, supported Moodle versions, installation instructions, and screenshots where relevant.

Required action: Move README from ../ to /plugin directory.

4.5 Missing CHANGES.md / CHANGELOG

A change log file is required for moodle.org plugin submissions to document version history.

5. What Is Implemented Correctly

The following areas meet Moodle standards and require no changes.

Category	Finding
Security	<code>require_capability()</code> checks present throughout all relevant pages
Security	<code>required_param()</code> / <code>optional_param()</code> used consistently — no direct <code>\$_GET</code> / <code>\$_POST</code> access (except <code>\$_FILES</code> noted above)
Security	<code>require_sesskey()</code> / <code>confirm_sesskey()</code> used for all write operations (except export action)
Security	<code>defined('MOODLE_INTERNAL')</code> <code>die()</code> present in all internal PHP files
Data access	Moodle DB API used consistently; no raw SQL queries

Category	Finding
Privacy	Privacy API implemented (provider.php)
Backup/Restore	Backup and restore class implemented
Code quality	PHPDoc annotations present on all classes and methods
i18n	German language strings (lang/de/) present alongside English
Assets	Plugin icon (pix/icon.svg) present

6. Recommended Fix Order

The following table lists all issues in recommended resolution order for submission readiness.

#	File / Location	Issue	Severity
1	edit_chart.php:145	Direct access to \$_FILES superglobal — violates Moodle Security Guidelines	CRITICAL
2	version.php	MATURITY_ALPHA — not acceptable for moodle.org/plugins production listing	CRITICAL
3	version.php	Incorrect \$plugin->requires (Moodle 3.10 value instead of Moodle 4.1+)	CRITICAL
4	amd/build/	Compiled AMD build directory missing — AMD modules will not load on production	CRITICAL
5	All PHP/JS files	License headers use http:// instead of https:// (20+ files)	MAJOR
6	All PHP/JS files	Non-standard license header — should reference Moodle, not plugin name	MAJOR
7	edit_chart.php:170	Export action missing sesskey verification — CSRF vulnerability	MAJOR
8	lib.php	File is empty — must either implement qtype_buchungssatz_pluginfile() or be removed	MAJOR
9	manage_charts.php, edit_chart.php	Use of die; instead of exit — violates Moodle Coding Style	MINOR
10	settings.php	File is empty — creates a blank settings page; should be removed	MINOR
11	amd/src/	AMD modules use legacy define() + jQuery; ES modules preferred for Moodle 4.x	MINOR
12	— (missing)	README file not in plugin directory — required for moodle.org plugin submission	MINOR
13	— (missing)	No CHANGES.md / CHANGELOG — required for moodle.org plugin submission	MINOR